

Lincoln University College

Faculty of Computer Science and Multimedia

Timed - A Medication Reminder Application

A PROJECT REPORT

Submitted to

Department of Information Technology

Phoenix College of Management

Maitidevi, Kathmandu

In partial fulfillment of the requirements for the Bachelor in Information
Technology (BIT)

Submitted by

Reezan Shrestha

BIT 5th Semester

University ID : LC0003001325

2024/12/26

DECLARATION

I hereby declare that the project report entitled "Timed - A Medication Reminder Application" is my original work, completed under the guidance of Mr. Saishab Bhattarai. This report has not been submitted elsewhere, in full or in part, for the award of any degree, diploma, or other academic recognition.

Signature:

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Lincoln University College
Faculty of Computer Science and Multimedia
Phoenix College of Management
Maitidevi, Kathmandu
Bachelors in Information Technology (BIT)

LETTER OF APPROVAL

The Project report entitled "**Timed - A Medication Reminder Application**" submitted by **Reezan Shrestha** , in partial fulfillment of the requirements for the Bachelor's degree in Information Technology, has been accepted as a record of work independently carried out by the group of departments.

In our opinion, it is satisfactory in the score and quality of a project for the required degree.

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ABSTRACT

The "Timed - Medication Reminder Application" is designed to enhance medication adherence and health management for individuals with chronic conditions such as hypertension and diabetes. By providing timely reminders for medications and medical appointments, the app helps prevent complications and improve overall health outcomes. It simplifies complex medication schedules, empowering users to manage their treatments independently and effectively, thereby improving quality of life. With customization options to cater to diverse patient needs, "Timed - Medication Reminder Application" extends its impact to broader public health systems, promoting healthier lifestyles and empowered healthcare management.

Keywords: Medication Reminder, Smartphone App, Appointment Scheduling, Digital Health, Personalized Care.

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Additionally, I extend my gratitude to the External Advisors for their encouragement and assistance during the course of this work. Lastly, I am profoundly grateful to my family and friends for their constant support and motivation, which helped me persevere and complete this project successfully.

CERTIFICATE

This is to certify that the project report entitled "Timed - A Medication Reminder Application" submitted by Reezan Shrestha University ID: LC0003001325 to Lincoln University College in partial fulfillment of the requirements for the Bachelor in Information Technology, is a record of original work carried out under my supervision and guidance.

Supervisor Name: Saishab Bhattarai

Designation: Project Coordinator

Signature:

Contents

LIST OF FIGURES	2
LIST OF TABLES	3
LIST OF ABBREVIATIONS	4
1 INTRODUCTION	5
1.1 Introduction	5
1.2 Problem Statement	6
1.3 Project Objectives	6
1.4 Significance of Project	6
1.5 Limitations of the Project	7
1.6 Development Methodology	7
1.7 Organization of the Report	8
2 Background Study and Literature Review	10
2.1 Background of the Study	10
2.2 Literature Review	11

3	System Analysis	13
3.1	System Analysis	13
3.1.1	Requirement Analysis	14
3.1.2	Feasibility Analysis	16
3.1.3	System Modeling	19
3.1.4	Dynamic modeling using State and Sequence Diagrams	21
3.1.5	Process modeling using Activity Diagrams	21
4	System Design	23
4.1	Refinement of Classes and Object	23
4.2	Component Diagram	24
4.3	Deployment diagram	27
5	Implementation and Testing	29
5.1	Implementation	29
5.1.1	Tools Used	29
5.1.2	Implementation Details of Modules	30
5.2	Testing	31
5.2.1	Test Cases for Unit Testing	31
5.2.2	Test Cases for System Testing	34
5.3	Result Analysis	35
6	Conclusion and Future Recommendations	36

6.1	Conclusion	36
6.2	Future Recommendations	37
	References	38
	Appendices	40
6.3	Firebase Schema	40
6.4	Design Mockups	41

List of Figures

3.1	Scrum Methodology	14
3.2	Gantt Chart	18
3.3	Use Case Diagram of Medication Reminder Application	19
3.4	class Diagram of Medication Reminder Application	20
3.5	Sequence Diagram of Reminder	21
3.6	Activity Diagram of Medication Reminder Application	22
4.1	class Diagram of Medication Reminder Application	24
4.2	Component diagram of A Medication Reminder Application	26
4.3	Deployment diagram of A Medication Reminder Application	28
6.1	Home Page	41
6.2	Landing Page	41
6.3	Profile Page	41

List of Tables

3.1	Schedule of "Timed - A Medication Reminder Application"	18
5.2	Test Cases for Unit Testing	33
5.4	Test Cases for System Testing	34

LIST OF ABBREVIATIONS

API: Application Programming Interface

ELA: Elderly Living Alone

EMR: Electronic Medical Records

ER: Entity-Relationship

EHR: Electronic Health Records

FCM: Firebase Cloud Messaging

iOS: iPhone Operating System

IT: Information Technology

mHealth: Mobile Health

RFID: Radio-Frequency Identification

RESTful API: Representational State Transfer (REST) Application Programming Interface

SMRAs: Smartphone Medication Reminder Applications

UAT: User Acceptance Testing

UI: User Interface

UML: Unified Modeling Language

UX: User Experience

Chapter 1

INTRODUCTION

1.1 Introduction

In the current era of rapid changes, one must be aware towards chronic clinical conditions, especially chronic hyper tension and diabetes, as they require regular follow up, drug schedule adherence, and visits to doctors. Most patients are unable to follow up on the typically busy treatment schedules, leading to a lower medication consumption level and causing complications, thereby increasing needless hospital admissions. Therefore, we have developed Timed- Medication Reminder Application, an all-in one solution to make medication adherence and health management with chronic diseases more efficient.

Timed- Medication Reminder Application keeps reminding users on time regarding medicines and medical appointments, thereby keeping the patients on track with their treatment plans. The user-friendly nature of the app and personal notifications attached to the app would thus serve as an agent enforcing patient adherence to prescribed therapies, resulting in better health outcomes, fewer emergencies, and an enhanced quality of life for the patients. It is therefore a worthy investment aimed at alleviating complicated drug schedules and making adherence autonomic so that it can not only lift the burden on health care systems.

1.2 Problem Statement

The problem of forgetting to take medications or just difficulties in management of complex regimens makes many middle-aged and older patients suffering from chronic conditions such as hypertension and diabetes not to adhere to their medications at all. As a result they experience worsened health outcomes with increased healthcare expenses[5]. We are in need of a user-friendly means by which to remind patients about times when they should take their medicine or when they have appointments scheduled. Thus, we designed our “Timed-Medicine Reminder Mobile Application” which sends timely reminders and allows tracking that enhance adherence to medications and improve health management.

1.3 Project Objectives

- To implement a timely notification system that consistently alerts users to take their medications and schedule upcoming doctor appointments.
- To effectively arrange and handle users prescription regimens and doctor appointments.
- To store and alert users with their medication and health data.

1.4 Significance of Project

The app called “Timed- Medication Reminder Application” is an important project that is meant to enhance medication adherence and health management for people suffering from chronic diseases like hypertension or diabetes. The timely reminders prompt users to take their medications as prescribed and attend their medical appointments accordingly. This way of following the prescriptions prevents complications and ultimately ensures better health outcomes. Besides, by making complex medication schedules easier to understand, this application enhances patient autonomy in medicine therapy i.e., they can administer treatment plans more efficiently. By

so doing, it leads not only to improved quality of life. Moreover, this project could be customized to cater to other patients thereby increasing its positive impact on public health systems. By doing so, we say that the “Timed-Medicine reminder application” fills a gap in health care as far as adherence is concerned thus promoting healthier and empowered citizens.

1.5 Limitations of the Project

- **Technology Reachability Compromised:** Elderly people may not possess smartphones or experience issues that limit their ability to utilize mobile applications, thus hampering the proper delivery of messages to the target audience by the app.
- **Data Safety Related Aspects:** A bigger question that arises from keeping health-related details is privacy as well as security. Organizations like health services must obey laws on data protection even if it may be tricky at times.
- **Limited Multi-language Support:** The application currently supports only the English language, limiting its usability for non-English speaking users. Further localization features can be incorporated to support multiple languages for broader accessibility in the future.
- **Cater only two diseases:** The application caters only hypertension and diabetes. It does not yet support a wide range of chronic conditions, which may limit its applicability for patients with other health issues. Future updates could expand the range of supported diseases to offer broader healthcare support.

1.6 Development Methodology

- **Scrum Methodology:** The "Timed - A Medication Reminder Application" follows the Scrum methodology, focusing on iterative development with regular sprint planning, daily stand-ups, and sprint reviews. This approach ensures

continuous collaboration, quick adaptation to changes, and the delivery of functional increments that meet user needs.

1.7 Organization of the Report

This report is structured into several chapters with each discussing a specific area of the project ‘*Timed - A Medication Reminder Application*’. The layout is as follows:

- **Chapter 1: Introduction**

This chapter presents an overview of the project including the introduction, problem statement, objectives of the project, significance, limitations encountered, and an outline of the report organization.

- **Chapter 2: Background Study and Literature Review**

This chapter brings together background of the study and all available scholarly work and published materials regarding treatment compliance, mHealth applications, and their importance in the management of long-term illnesses, while also pointing out the gaps in the present solutions that this project seeks to address.

- **Chapter 3: System Analysis**

The System Analysis section provides a comprehensive overview of the project’s requirements, objectives, and constraints. It outlines the functional and non-functional requirements, feasibility studies, and the methodologies used to gather information, ensuring a clear understanding of the system’s purpose and the processes involved in its development.

- **Chapter 4: System Design**

The System Design section details the architecture and components of the system, including both high-level and detailed designs. It outlines the design decisions, models, and diagrams used to structure the system, ensuring that it meets the specified requirements and is scalable, efficient, and user-friendly.

- **Chapter 5: Implementation and Testing**

The Implementation and Testing section describes the process of translating the system design into a working software application. It covers the coding, integration, and deployment stages, as well as the testing methodologies employed to ensure the system functions correctly, meets user requirements, and is free from defects. This section also highlights the types of tests conducted, such as unit testing, system testing, and user acceptance testing.

- **Chapter 6: Conclusion and Recommendations**

This final chapter contains a summary of findings, conclusions drawn from the project, and suggestions for future improvements or developments.

- **Bibliography**

This section includes a list of all sources and references mentioned throughout the report, formatted in the appropriate citation style.

- **Appendices**

This section contains additional information, such as technical documents, screenshots of the app, or any other materials that support the project but are not part of the main chapters.

Chapter 2

Background Study and Literature Review

2.1 Background of the Study

In particular, older people and midlife patients suffering from chronic ailments like hypertension and diabetes mellitus often forget their prescriptions as prescribed. Per the research report of S. Balakrishnan between July and September 2020, a community-based cross-section survey was carried out in three districts of Eastern Nepal. The findings revealed that over half (66.5%) of the elderly (60 years and above) had at least one chronic non-communicable disease including diabetes (13.5%) or hypertension (31.6%)[3]. This group faces challenging times in maintaining good health management because of the high occurrences.

In addition, complicated medication schedules and frequent lapses on taking medication have resulted into non adherence with the involvement of poorer health outcomes hence relying more on hospitals. This problems can be solved through the use of smartphone medication reminder applications (SMRAs), which are affordable and realistic steps forward[7]. These applications utilize a new method to increase adherence by sending notifications for medication as well as doctor appointments in such a timely manner.

Our Flutter-based project, the "Timed- Medication Reminder Application", is designed to have customized reminders and user-friendly UI such that it may enhance the clarity of patients understanding of how medication should be taken. The novel approach for managing adherence challenges is smartphone adherence applications which provide state of the art functionalities directly to patients and healthcare givers in order to facilitate better drug taking behavior at their level.[4] This project intends to exploit mobile technologies to ease the burden on health systems and improve health outcomes.

2.2 Literature Review

Medication reminder systems, According to Deepti Ameta, My MediHealth use mobile devices for the management and reminders of medication schedule particularly for children. More advanced systems utilize RFID technology and sensors to ensure adherence. However, issues regarding functionality and user participation persist[2]. Research has shown that for senior citizens, Elderly Living Alone (ELA) missing doses is often caused by forgetting to take medicines or having no one to remind them about those schedules if they live alone. Most people aged 60 years and above face challenges when it comes to remembering and monitoring time for drug administration.[11].

As outlined in B. Aggarwal, After 30 days of follow-up, only 1 out of 10 individuals claimed to have been admitted back to the hospital in the last month but none of them adhered to medication as expected. On average, they missed one or more pills or doses each week. There was an inclination that baseline depression scores for those who did not take their medications at all during this period exceeded by much those who did take them[1]. Numerous elements of the Wedjat app contribute to a decrease in medication mistakes. It not only notifies users when to take their medications on time, but it also alerts them to potential drug interactions. Furthermore, calendars are used to send reminders and automatically reschedule missed doses. Through synchronizing with EMR systems, this app can maintain a record of the medications you take[10]. According to Santo nonetheless, it is important to emphasize that the

effectiveness of these applications should be subjected to further studies with larger samples as well as longer follow-up periods[9]

Based on the findings presented in Prafful Naikode, Medsy is yet another piece of software that aims to develop a medication reminder system. Someone would be really irate, for instance, if they had to take their medication three times a day and this app would not let them set an alarm for it[6]. We attempted to address this issue in a unique manner by letting platform users set up as many notifications and reminders as they would like. As outlined in Ms. S Medication reminder systems is a life-saving, money-saving and time-saving software that is straightforward to apply and offers an exquisite person interface[8].

Chapter 3

System Analysis

3.1 System Analysis

The "Timed - A Medication Reminder Application" is developed using the Scrum methodology, which involves a series of iterative processes. Initially, the requirement analysis phase is conducted to gather and prioritize both functional and non-functional requirements. Based on these requirements, the team creates a product backlog, which guides the development process. The system is designed incrementally in each sprint, with the design evolving as new features are developed. During each sprint, the development team codes and implements the features, ensuring that each iteration results in a potentially shippable product increment.

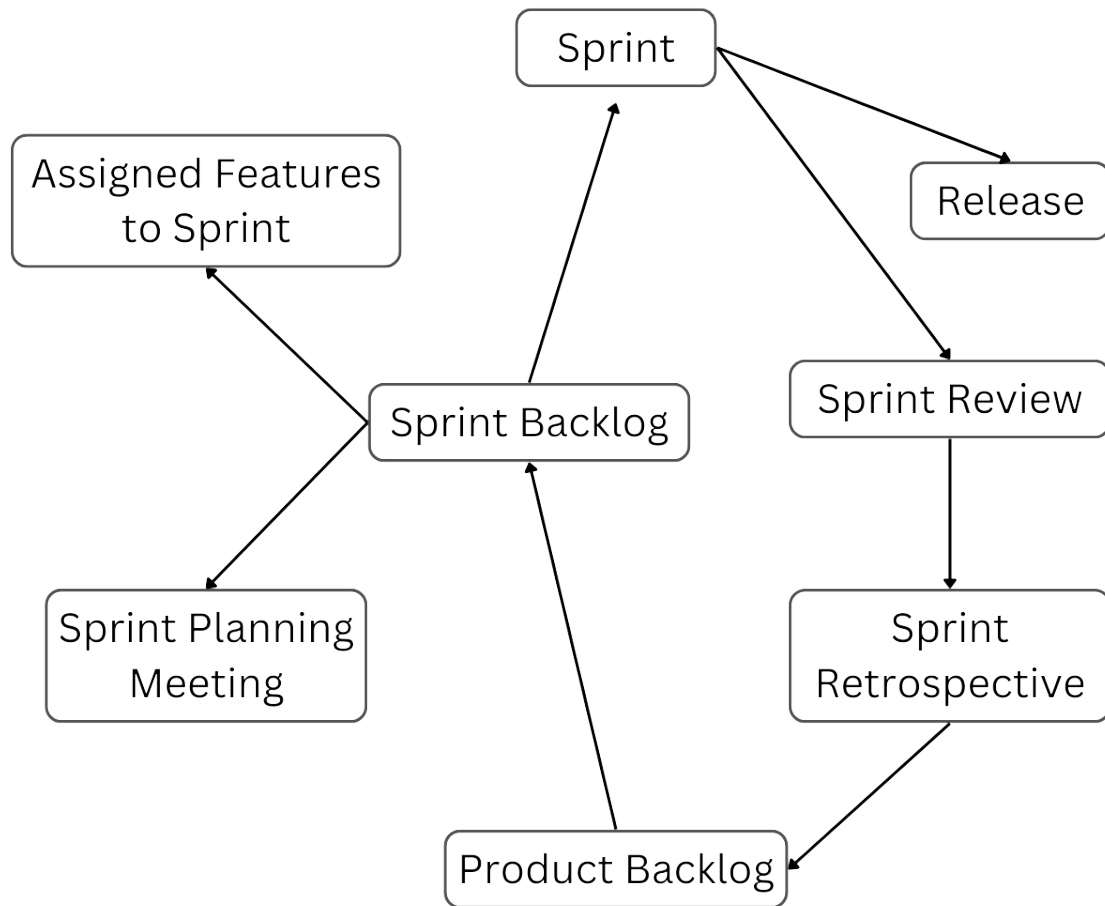


Figure 3.1: Scrum Methodology

3.1.1 Requirement Analysis

Before commencing the development of "Timed - A Medication Reminder Application", requirement gathering research was carried out extensively, as the program was intended to be functional and non-functional requirements.

Functional Requirements

The "Timed - A Medication Reminder Application" includes the following key functional requirements.

1. **User Login:** Users must be able to securely log in to their accounts to access personalized medication schedules and settings.

2. **Register Medicine Data:** Users should be able to register detailed information about their medications, including the name, time and date of the drug. This helps ensure that users are organized and on track with their prescribed treatment.
3. **Receive Medication and Dosage Reminders:** The application must send timely reminders to users that they should take the medications at the correct times and administer the correct dose. Users can choose to snooze or dismiss these reminders according to their preferences.
4. **Appointment Scheduling and Reminders:** Users can schedule medical appointments for check-ups or follow-ups within the app. The system will send timely reminders about upcoming appointments, ensuring that users don't miss their scheduled visits.
5. **Medicine Supply Notifications:** The app will alert users when their medication supply is running low. This feature helps users stay ahead of their medication needs and avoid running out of critical prescriptions.

Non-Functional Requirements

Various non-functional requirements have been analyzed and identified, which are outlined as follows:

- **Performance:** The app should send reminders quickly and handle many users at the same time without delays.
- **Usability:** The app should be easy to use, letting users set up their medication reminders quickly. It should also support features for people with accessibility needs, like screen readers.
- **Reliability:** The app should be available most of the time and include features to fix errors or recover from issues.
- **Security:** User data should be kept safe and private, using encryption and secure communication.
- **Scalability:** The app should be able to grow and handle more users as needed.
- **Compatibility:** The app should work well on both Android and iOS devices, including those with lower performance.

3.1.2 Feasibility Analysis

To ensure the successful development and implementation of the "Timed - A Medication Reminder Application," a thorough feasibility study was conducted. The study evaluated the project's viability from technical, operational, economic, and scheduling perspectives, as outlined below:

Technical Feasibility Study

The project leverages modern, accessible technologies to ensure technical feasibility. **Flutter** is used for cross-platform development, providing seamless performance on both Android and iOS platforms. **Firestore** serves as the backend, offering reliable

cloud-based database storage and authentication services. The required programming tools, frameworks, and libraries are readily available and well-documented, ensuring that the technical requirements of the application can be met efficiently.

Operational Feasibility Study

The application is designed to be user-centric, focusing on simplicity and ease of use. Its intuitive **User Interface (UI)** allows users to manage medication schedules and appointments effortlessly, requiring only basic familiarity with smartphones. The system ensures accessibility for users with diverse technical skills, making it practical for patients managing chronic health conditions.

Economic Feasibility Study

Cost-effectiveness is a key factor in the project's economic feasibility. By utilizing open-source tools like **Flutter** and **Firestore**, development costs are minimized while maintaining high-quality output. The system does not require any additional hardware, as it operates seamlessly on existing smartphones. This approach reduces overhead costs and ensures affordability for both developers and end-users.

Schedule Feasibility Study

The project has been planned meticulously to ensure timely completion within the scheduled timeframe. Each development phase, from research to deployment, is assigned specific durations to maintain focus and efficiency, reducing the risk of delays.

Task Name	Duration
Planning	1 weeks
Requirement Analysis	2 weeks
Design	5 weeks
Development	11 weeks
Testing and Debugging	13 weeks

Table 3.1: Schedule of "Timed - A Medication Reminder Application"

GANTT CHART

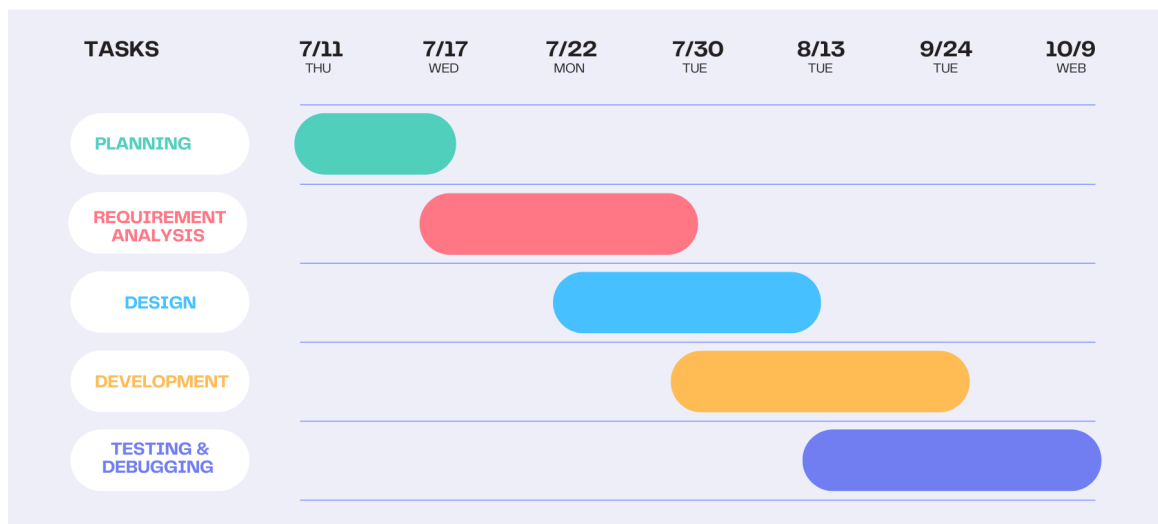


Figure 3.2: Gantt Chart

3.1.3 System Modeling

The process of developing abstract models of a system, with each model presenting a different view or perspective of that system.

- **Use Case Diagram:**

A Use Case Diagram is a visual representation of the interactions between users (or actors) and a system.

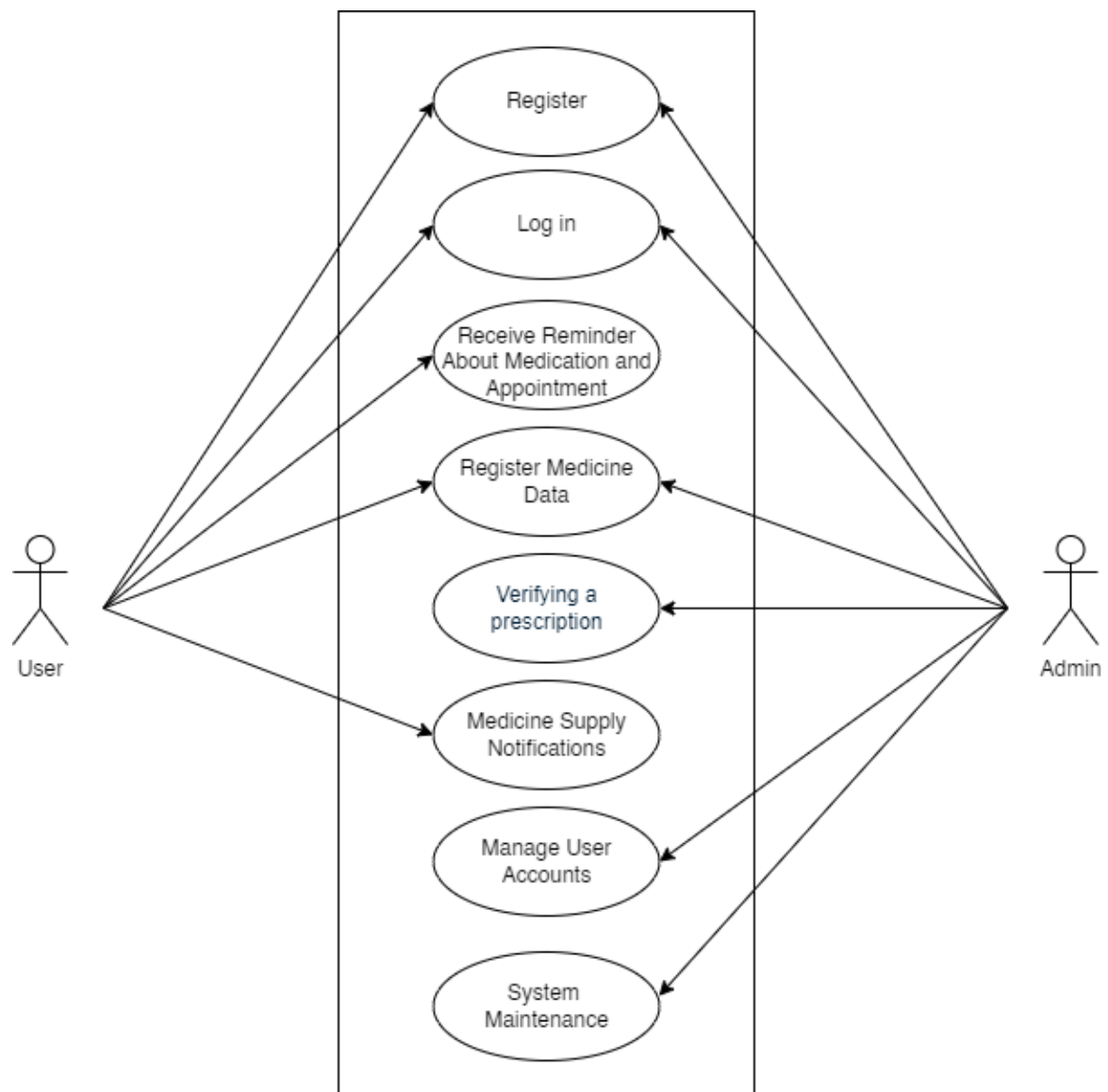


Figure 3.3: Use Case Diagram of Medication Reminder Application

- **Class Diagram:**

A Class Diagram shows the system's classes, their attributes, methods, and the relationships between them. It provides an overview of how the system is organized and how objects interact.

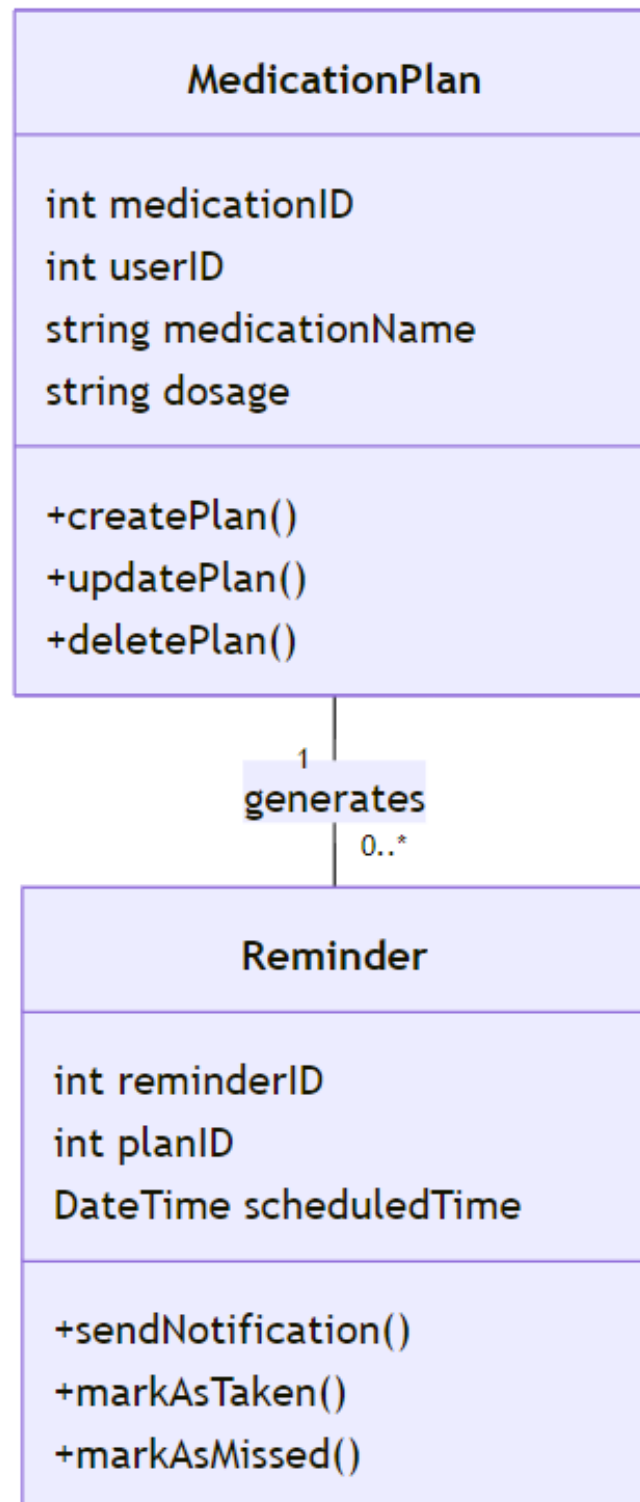


Figure 3.4: class Diagram of Medication Reminder Application

3.1.4 Dynamic modeling using State and Sequence Diagrams

The sequence diagram illustrates the interactions between system components, helping to visualize both current behavior and the requirements for new features.

- **Sequence Diagram**

sequence diagram visually represents how objects or components in the system interact over time. It helps to understand the flow of processes and the order in which operations occur. For the "Timed - A Medication Reminder Application," the sequence diagram would detail the sequence of actions between the user, the medication reminder system, and other system components.

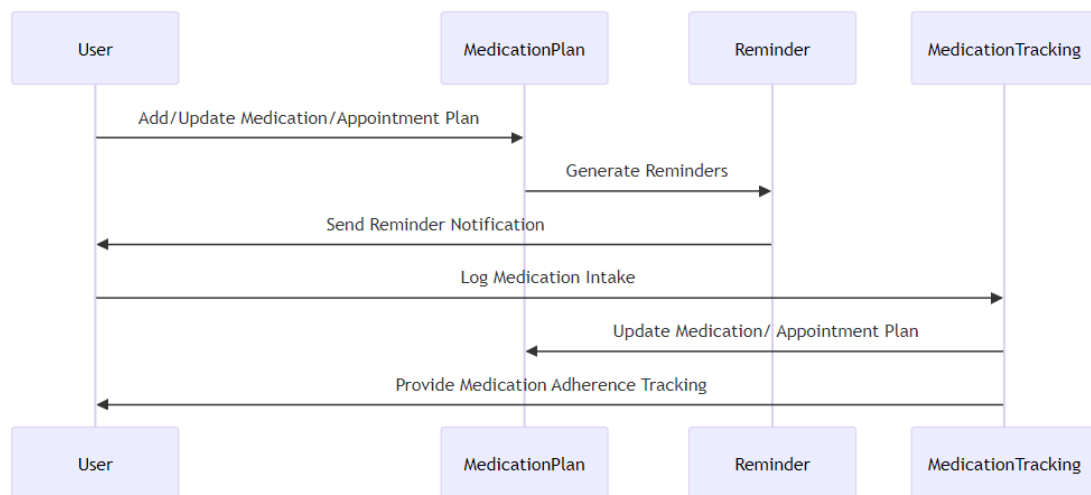


Figure 3.5: Sequence Diagram of Reminder

3.1.5 Process modeling using Activity Diagrams

- **Activity Diagram**

An activity diagram represents the workflow of the system, showcasing the sequence of activities and decisions involved.

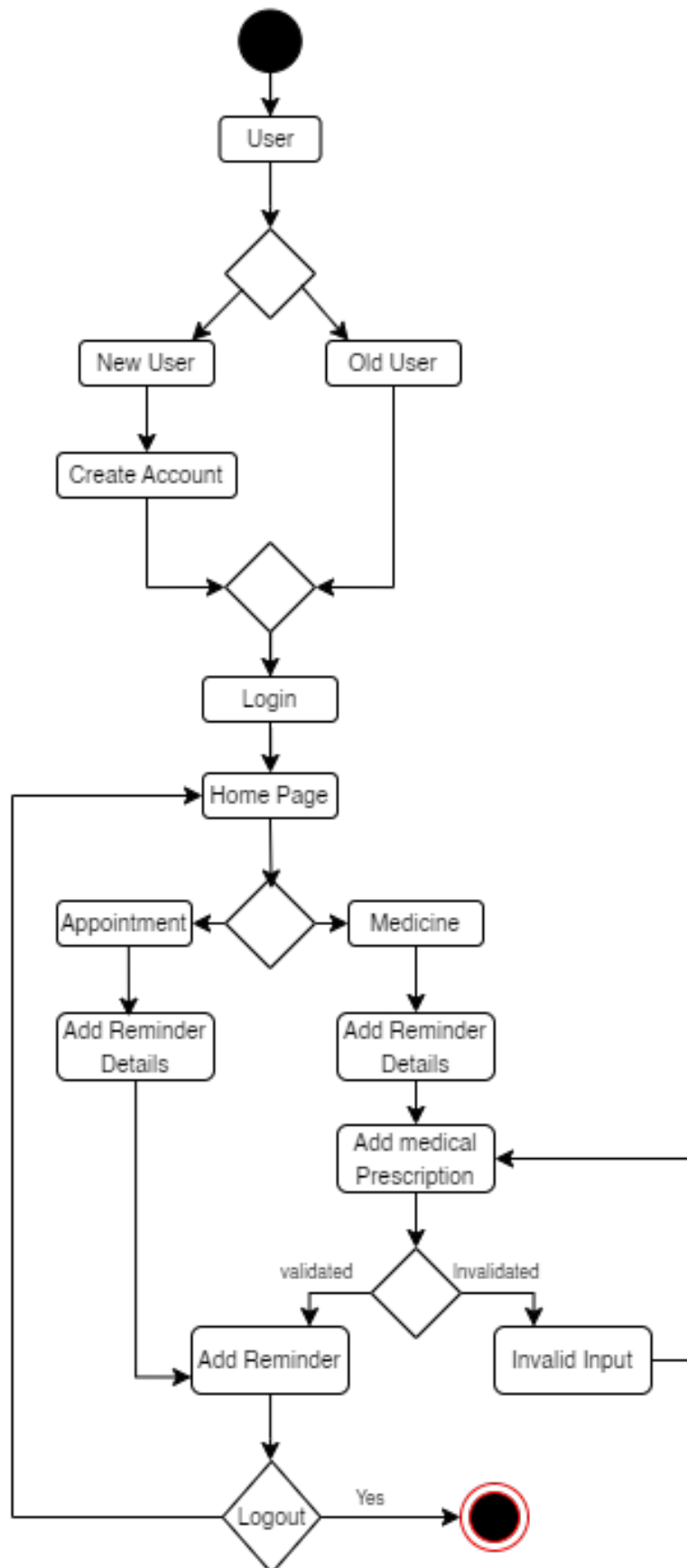


Figure 3.6: Activity Diagram of Medication Reminder Application

Chapter 4

System Design

System design is a crucial phase in the software development process that involves defining the architecture, components, modules, and data flow of the system. It transforms the requirements gathered during the analysis phase into a blueprint for constructing the system. The design phase ensures that the system will be both effective and efficient in meeting its functional and non-functional requirements. It outlines how the system will be structured, how its components will interact, and how the data will flow throughout the system.

4.1 Refinement of Classes and Object

Initially, the identification and further refining of the essential classes will be considered in the object-oriented system design. These classes mainly represent entities of Timed - A Medication Reminder Application. They are designed from the functional requirements of the system.

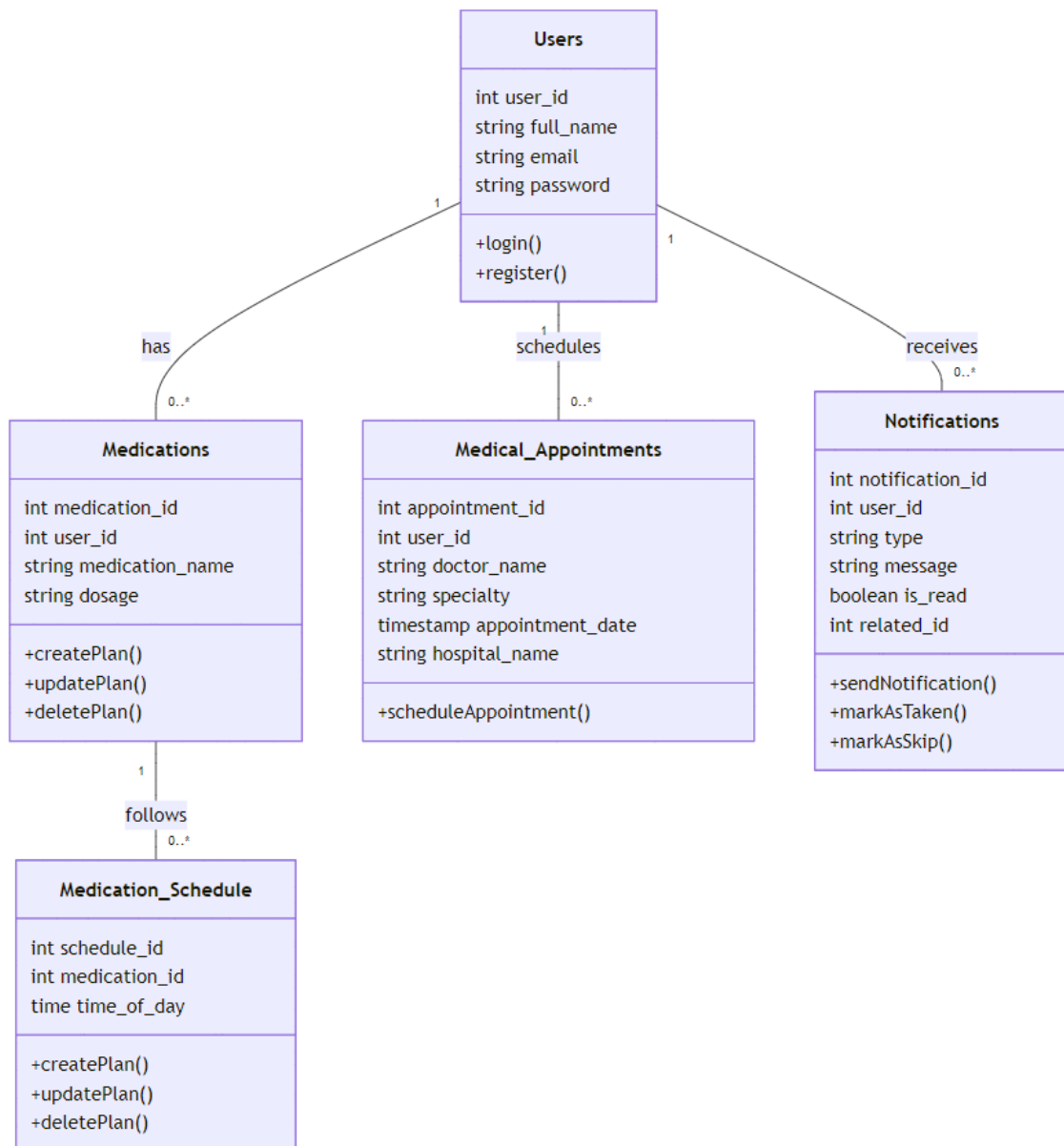


Figure 4.1: class Diagram of Medication Reminder Application

4.2 Component Diagram

The Component Diagram illustrates the relationship between the various components of the system and how they interact with one another. It shows the high-level view of the system's architecture from an object-oriented perspective.

i. Frontend Component:

Built with Flutter, this component provides an intuitive interface for managing reminders, viewing medication schedules, and handling notifications. It manages user interactions and communicates with the backend for data synchronization.

ii. Backend Component:

Implemented in Flutter, it handles the core functionalities like scheduling reminders, sending notifications, and processing user input locally.

iii. Database Component:

The **Firestore Realtime Database**, hosted in the cloud, handles all data storage and retrieval operations. The database ensures **real-time synchronization** between user devices, providing secure and efficient data management.

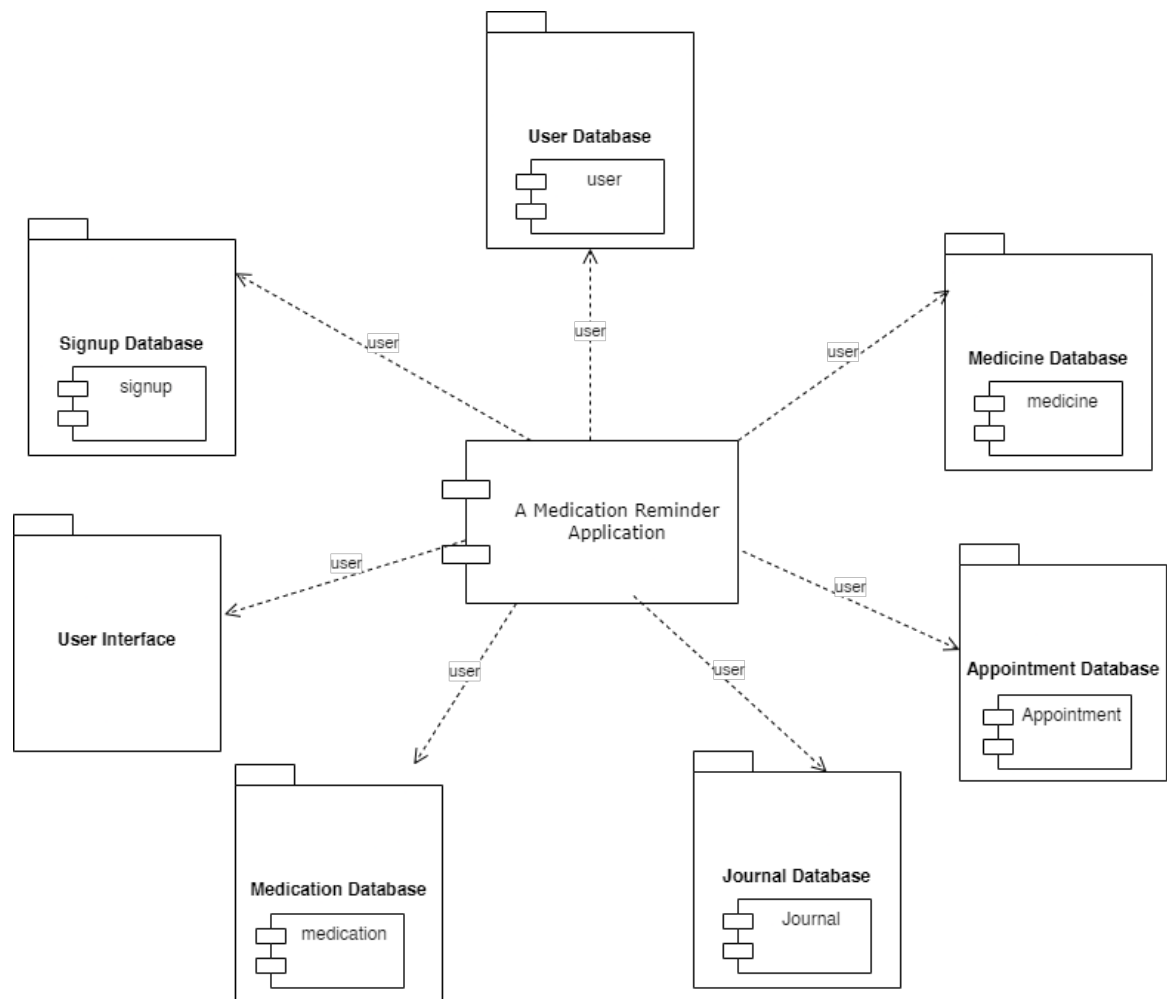


Figure 4.2: Component diagram of A Medication Reminder Application

4.3 Deployment diagram

The Deployment Diagram describes the physical deployment of the system's components on hardware resources. It shows how the system's software is distributed across various hardware nodes and how these nodes communicate with each other.

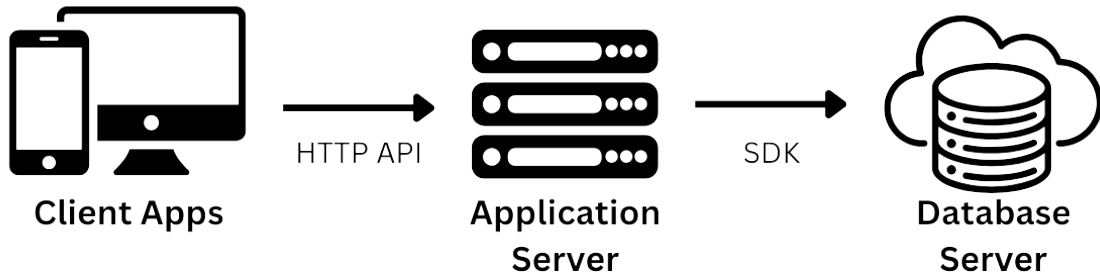
i. User Devices (Client-Side):

Users will access the Timed - A Medication Reminder application via their smartphones. The frontend and backend of the application run on these devices, developed using Flutter, providing an intuitive interface and managing all core functionalities.

ii. Database Server:

The Firebase Realtime Database is hosted in the cloud and handles all data storage and retrieval operations. It securely stores user profiles, medication schedules, and notification settings, ensuring real-time synchronization between user devices and the database.

Traditional



Firestore

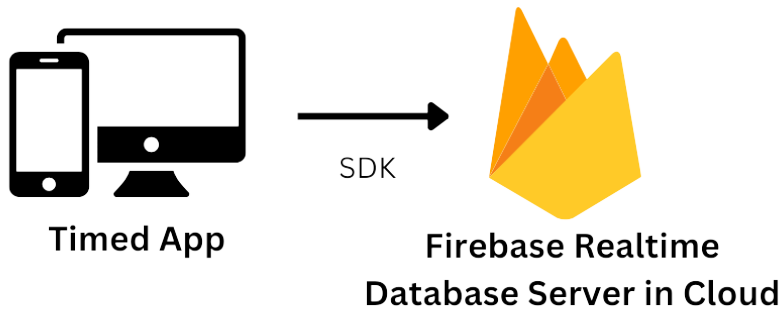


Figure 4.3: Deployment diagram of A Medication Reminder Application

Chapter 5

Implementation and Testing

5.1 Implementation

The implementation of the Timed - A Medication Reminder Application project was carried out using a combination of integrated technologies for the user interface, back-end logic, and a robust cloud-based database system. This section outlines the tools, technologies, and modules utilized to develop the application, including a detailed explanation of the implementation process for each module.

5.1.1 Tools Used

The following tools were used for the development and implementation of the "Timed-A Medication Reminder Application":

- **Flutter:** A cross-platform framework used to build the mobile app for both iOS and Android platforms.
- **Dart:** The programming language used with Flutter for writing the app's business logic.
- **Firebase:** Provides backend services, including a real-time database for storing user data and cloud functions for handling notifications.
- **Firebase Cloud Messaging (FCM):** Used for sending push notifications to users for reminders on medication and appointments.
- **GitHub:** For version control and collaboration among the development team.

- **Android Emulator / iOS Simulator:** Used for testing the app across different device configurations.
- **Trello:** A project management tool used for organizing tasks, tracking progress, and collaborating with the development team.
- **Figma:** A design tool used for creating user interface (UI) mockups and visual designs for the app.

5.1.2 Implementation Details of Modules

The "Timed" app is divided into several modules to ensure its scalability and maintainability:

- **User Module:** Manages user information, including medical history and medication schedules.
- **Medication Module:** Defines medication details, including name, dosage, and frequency of intake.
- **Reminder Module:** Responsible for scheduling and sending notifications to the user at the correct times for medication and appointments.
- **Appointment Module:** Stores details of medical appointments, including the doctor's name, date, time, and reason for the appointment.
- **Health Journal Module :** The Health Journal Module allows users to record their daily health data for future reference, including blood pressure and sugar levels.
- **Schedule Module:** Manages the scheduling of medications and appointments, triggering reminders at appropriate times.

5.2 Testing

Testing was performed to verify the functionality, usability, and performance of the Timed - A Medication Reminder Application. The following testing methodologies were employed:

5.2.1 Test Cases for Unit Testing

Unit testing happened to ensure correctness in single parts or components of the system. For each function, the expected output and edge-case handling were tested.

SN	Test Case	Input Data	Expected Outcome	Actual Output	Remarks
1	Login Page	Full Name: testing and password: testing	Invalid username or password	Invalid username or password	PASS
2	Register	Full Name: testing Email: testgmail.com Password: test@T123 Repassword: test@T123	Invalid email address!	Invalid email address!	PASS
3	Register	Full Name: testing Email: test@gmail.com Password: test Repassword: test	Weak password, please enter a strong password	Weak password, please enter a strong password	PASS

4	Register	Full Name: testing Email: test@gmail.com Password: test@T123 Repassword: test@T123	Account created	Account created	PASS
5	Login Page	Email: test@gmail.com Password: test@T123	Redirect to the Home page	Redirect to the Home page	PASS
6	Medicine Add Page	Unit: Mylod5 How Often: Once daily Time: 8:00 AM Quantity: 5 pill(s)	Medication reminder added suc- cessfully	Medication reminder added suc- cessfully	PASS
7	Add Ap- pointment	Hospital Name: om hospital Appointment Name: eye check Date: Dec 1, 2024	Reminder added suc- cessfully	Reminder added suc- cessfully	PASS
8	Change Password	Current Password: test@T123 New Password: TEST@t1234 Confirm Password: TEST@t1234	change password success- fully	change password success- fully	PASS
9	Home	Take medicine	Number of medicine decrease	Number of medicine decrease	PASS
10	Logout	Are you sure you want to log out: Logout	Redirect to the Login Page	Redirect to the Login Page	PASS

11	Logout	Are you sure you want to log out: Cancel	Redirect to the Profile Page	Redirect to the Profile Page	PASS
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Table 5.2: Test Cases for Unit Testing

5.2.2 Test Cases for System Testing

The testing of an entire system was done with evaluating the full system by determining if all the components are functional together.

SN	Types of Test	Description
1	End-to-End Test	Simulate the entire user journey from signing up for the service, setting up medication reminders, viewing reminders, and being given notifications. The user was able to register, set reminders, and receive notifications without errors.
2	Performance Test	Evaluate how quickly the system can respond to multiple users setting and sending reminders at the same time. The system performed well with minimal latency.
3	Notification Test	Test the reliability of sending medication and appointment reminders on time. All notifications were received accurately and on time.
4	Usability Test	Verify that the user interface is intuitive and user-friendly. The platform provided easy navigation, and accessibility for all users, including elderly patients.

Table 5.4: Test Cases for System Testing

5.3 Result Analysis

The testing phase revealed that the "Timed-A Medication Reminder Application" app functions as expected across all modules. Unit tests passed successfully for individual components such as medication scheduling, user data management, and appointment reminders. System tests confirmed that the app's modules interact seamlessly, with notifications triggered accurately and user data syncing properly between local and cloud databases. Performance testing showed that the app performs well on both Android and iOS devices, with minimal lag or delays in loading.

Overall, the app performed well during testing and is ready for deployment after minor optimizations based on user feedback.

Chapter 6

Conclusion and Future Recommendations

6.1 Conclusion

"Timed-A Medication Reminder Application" is aimed at improving the management of medication adherence among patients with chronic diseases such as diabetes and hypertension. Medication non-adherence is a significant problem affecting older adults and middle-aged patients across the world. By providing timely reminders for medication and medical appointments, "Timed-A Medication Reminder Application" empowers individuals to take control of their health. The easy-to-use interface helping to address a common issue and reducing hospital visits and associated costs. The app also enhances patient autonomy, enabling users to manage their medication schedules independently and efficiently.

However, despite its promising features, "Timed-A Medication Reminder Application" still faces several challenges. These include issues related to access to technology, user compliance, data security concerns, and device limitations. Not all users may have access to the necessary devices or reliable internet connections, particularly in underprivileged regions. Furthermore, ensuring that users remain consistent with their medication adherence requires continuous user engagement and motivation. Security concerns regarding user data, especially sensitive health information, must also be addressed through robust encryption and secure data storage mechanisms. Additionally, device limitations, such as varying operating system versions and screen sizes,

may affect the app's performance and user experience.

6.2 Future Recommendations

To improve the app and address these challenges, future work could focus on:

- **Improved User Engagement:** Incorporating features such as gamification, reminders for missed doses, and incentives for consistent medication adherence can motivate users to maintain their schedules.
- **Accessibility and Inclusivity:** Expanding the app's reach to users with limited access to smartphones or the internet by offering offline capabilities or compatibility with low-cost devices.
- **Data Security Enhancements:** Strengthening data protection by implementing end-to-end encryption and complying with healthcare data protection regulations to ensure user confidentiality.
- **Platform Optimization:** Ensuring the app is compatible with a wide range of devices and operating systems, providing a smooth user experience across different screen sizes and configurations.
- **Integration with Healthcare Systems:** Future versions of the app could integrate with electronic health records (EHR) systems, allowing healthcare providers to monitor patient adherence and make adjustments to treatment plans as needed.

In conclusion, while "Timed- A Medication Reminder Application" demonstrates significant potential to improve medication adherence and enhance the quality of life for patients with chronic conditions, continuous improvements and adjustments are necessary to make the app more accessible, secure, and engaging. With the right modifications, "Timed-A Medication Reminder Application" can achieve its goal of helping patients with hypertension, diabetes, and other chronic diseases manage their medication schedules more effectively, leading to better health outcomes and reduced healthcare costs.

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Appendices

6.3 Firebase Schema

Users Collection

The ‘users’ collection stores user account details.

- **Document Structure:**

```
{  
  "Full name": "ram rai",  
  "email address": "abc@example.com",  
  "password": "hashed_password",  
  "conform password": "hashed_password",  
}
```

- **Fields:**

- **full name:** Full name of the user.
- **email:** Email address of the user.
- **password:** Hashed password for authentication.
- **conform password:** conform Hashed password for authentication.

6.4 Design Mockups



Figure 6.1: Home Page



Figure 6.2: Landing Page

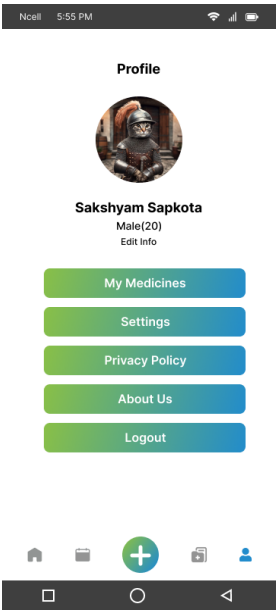


Figure 6.3: Profile Page